

PS3 Q3 HAMILTONIAN SYSTEM

$$\frac{dx}{dt} = y \quad (X(x, y))$$

$$\frac{dy}{dt} = 2 - x - 3x^2 \quad (Y(x, y))$$

HAMILTONIAN

$$\frac{\partial X}{\partial x} = 0, \quad \frac{\partial Y}{\partial y} = 0$$

$$\boxed{\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} = 0}$$

THE SYSTEM IS
HAMILTONIAN.

$$H(x, y) = \frac{\partial H}{\partial x} = -\frac{\partial y}{\partial t} \quad \wedge \quad \frac{\partial H}{\partial y} = \frac{\partial x}{\partial t}$$

$$\frac{\partial H}{\partial x} = -Y(x, y) = 3x^2 + x - 2 \Rightarrow H(x, y) = x^3 + \frac{x^2}{2} - 2x + g(y)$$

$$\frac{\partial H}{\partial y} = +X(x, y) = y \Rightarrow H(x, y) = \frac{y^2}{2} + f(x)$$

THUS $H(x, y) = x^3 + \frac{1}{2}(x^2 + y^2) - 2x + C$

BY CHOOSING $C=0$

$$\boxed{H(x, y) = x^3 + \frac{1}{2}(x^2 + y^2) - 2x}$$

DOUBLE CHECK

$$J = \begin{bmatrix} 0 & 1 \\ -1-6x & 0 \end{bmatrix} \quad x=0 \\ \Delta = 1+6x$$

E.P.

$$y=0 \quad \left(\begin{matrix} -1 \\ 0 \end{matrix} \right), \left(\begin{matrix} \frac{2}{3} \\ 0 \end{matrix} \right) \\ 2-x-3x^2=0$$

CLASSIFICATION

$$\left(\begin{matrix} -1 \\ 0 \end{matrix} \right) \Rightarrow x=0 \Rightarrow \Delta = -5 \Rightarrow \text{SADDLE POINT}$$

$$\left(\begin{matrix} \frac{2}{3} \\ 0 \end{matrix} \right) \Rightarrow x=\frac{2}{3} \Rightarrow \Delta = 5 \Rightarrow \text{CENTRE}$$

CLASSIFICATION

$$\frac{\partial H}{\partial x} = 0 \Rightarrow 3x^2 + x - 2 = 0 \Rightarrow x = -1 \vee x = \frac{2}{3}$$

$$\frac{\partial H}{\partial y} = 0 \Rightarrow y = 0$$

$$\text{E.P. ARE } \begin{pmatrix} -1 \\ 0 \end{pmatrix} \wedge \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix}$$

$$\Delta = \frac{\partial^2 H}{\partial x^2} \frac{\partial^2 H}{\partial y^2} - \frac{\partial^2 H}{\partial x \partial y} = (6x+1) \times 1 - 0 = 6x+1$$

$$\Delta(-1, 0) = -5 \quad \text{SADDLE POINT}$$

$$\Delta\left(\frac{2}{3}, 0\right) = 5 \quad \text{CENTRE}$$

PHASE PORTRAIT

We need to study the function $H(x, y) = x^3 + \frac{1}{2}(x^2 + y^2) - 2x$

$$\bullet H\left(\frac{2}{3}, 0\right) = \frac{8}{27} + \frac{2}{9} - \frac{4}{3} - \frac{4}{3} = \frac{8+6-36}{27} = -\frac{22}{27}$$

$$\bullet H(-1, 0) = -1 + \frac{1}{2} + 2 = \frac{3}{2}$$

• Asymptotic analysis

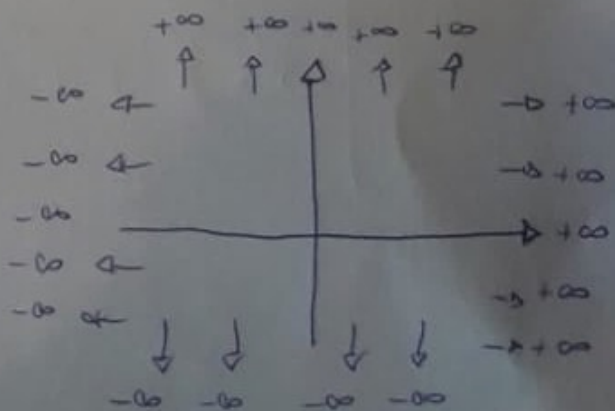
$$\lim_{x \rightarrow +\infty} H = +\infty \quad \text{for fixed } y$$

$$\lim_{x \rightarrow -\infty} H = -\infty \quad \text{for fixed } y$$

$$\lim_{y \rightarrow +\infty} H = +\infty \quad \text{for fixed } x$$

$$\lim_{y \rightarrow -\infty} H = -\infty \quad \text{for fixed } x$$

H assumes all values between $(-\infty, +\infty)$



LEVEL CURVES

We aim to find curves in the x, y plane such that

$$H(x, y) = C \quad \text{for } C \text{ constant.}$$

INTERESTING INTERVALS ARE

<u>I</u>	$C < -\frac{22}{7}$	H is less than in the centre point
<u>II</u>	$C = -\frac{22}{7}$	H is the same as in the centre point
<u>III</u>	$-\frac{22}{7} < C < \frac{3}{2}$	H assumes values between the one in the centre point and the one in the saddle point
<u>IV</u>	$C = \frac{3}{2}$	H is the the same as in the saddle point
<u>V</u>	$C > \frac{3}{2}$	H is greater than in the saddle point

For each of the five cases highlighted we are going to draw the level curves.

Finally, we will put everything together and sketch the PHASE PORTRAIT.

II $C = -\frac{22}{7}$

[It is convenient to start with the ~~center~~ level curve of the values at the e.p. It simplifies later work.]

$H = -\frac{22}{7}$ at the centre (obviously) and at the curve

$$x^3 + \frac{1}{2}(x^2 + y^2) - 2x = -\frac{22}{7}$$

therefore

$$y = \pm \sqrt{-2x^3 - x^2 + 4x - \frac{44}{7}}$$

WE NEED TO DRAW THIS.

• INTERSECTION WITH THE VERTICAL AXIS

$$x=0 \Rightarrow y = \pm \sqrt{-\frac{44}{7}}$$

There is no intersection with the vertical axis

• INTERSECTION WITH THE HORIZONTAL AXIS

$$y=0 \Rightarrow -2x^3 - x^2 + 4x - \frac{44}{7} = 0$$

Not too easy! We need to use Cardano's Method

$$-2x^3 - x^2 + 4x - \frac{44}{7} = 0$$

$$z = x + \frac{1}{6} \Rightarrow x = z - \frac{1}{6}$$

$$-2\left(z - \frac{1}{6}\right)^3 - \left(z - \frac{1}{6}\right)^2 + 4\left(z - \frac{1}{6}\right) - \frac{44}{7} = 0$$

$$-2\left(z^3 - \frac{z^2}{2} + \frac{z}{12} - \frac{1}{216}\right) - \left(z^2 - \frac{z}{3} + \frac{1}{36}\right) + 4\left(z - \frac{1}{6}\right) - \frac{44}{7} = 0$$

$$-2z^3 + z^2 - \frac{z}{6} - \frac{1}{108} - z^2 + \frac{z}{3} - \frac{1}{36} + 4z - \frac{2}{3} - \frac{44}{7} = 0$$

$$-2z^3 + \frac{25}{6}z - \frac{2635}{378} = 0$$

$$z^3 - \frac{25}{12}z + \frac{2635}{756} = 0$$

Let $P = -\frac{25}{12} \approx -2.083$

and $Q = \frac{2635}{756} \approx 3.485$

(Note that, for both u and v , there are also other two complex roots. However we don't need them.)

then $z = u + v$ where

$$u^3 = -\frac{Q}{2} + \sqrt{\frac{Q^2}{4} + \frac{P^3}{27}} \Rightarrow u^3 \approx -0.039 \Rightarrow u \approx -0.463$$

$$v^3 = -\frac{Q}{2} - \sqrt{\frac{Q^2}{4} + \frac{P^3}{27}} \Rightarrow v^3 \approx -3.386 \Rightarrow v \approx -1.502$$

HENCE

$$x = u + v + \frac{1}{6} = -0.463 - 1.502 - 0.167 = -2.132$$

VERIFY

$$-2(-2.132)^3 - (-2.132)^2 + 4(-2.132) - \frac{44}{7} \approx 0.022$$

THEREFORE

INTERSECTION WITH $y=0$ IS

$$x \approx -2.132$$

↑
CLOSE TO ZERO, WE ARE CLOSE TO THE ACTUAL ROOT (WHICH IS -2.1303 ACTUALLY)

LET US PICK SOME VALUES OF x TO DRAW THE CURVE LEVEL

• $x = -2 \Rightarrow y = \pm \sqrt{-\frac{16}{7}}$ THERE IS NO VALUE.

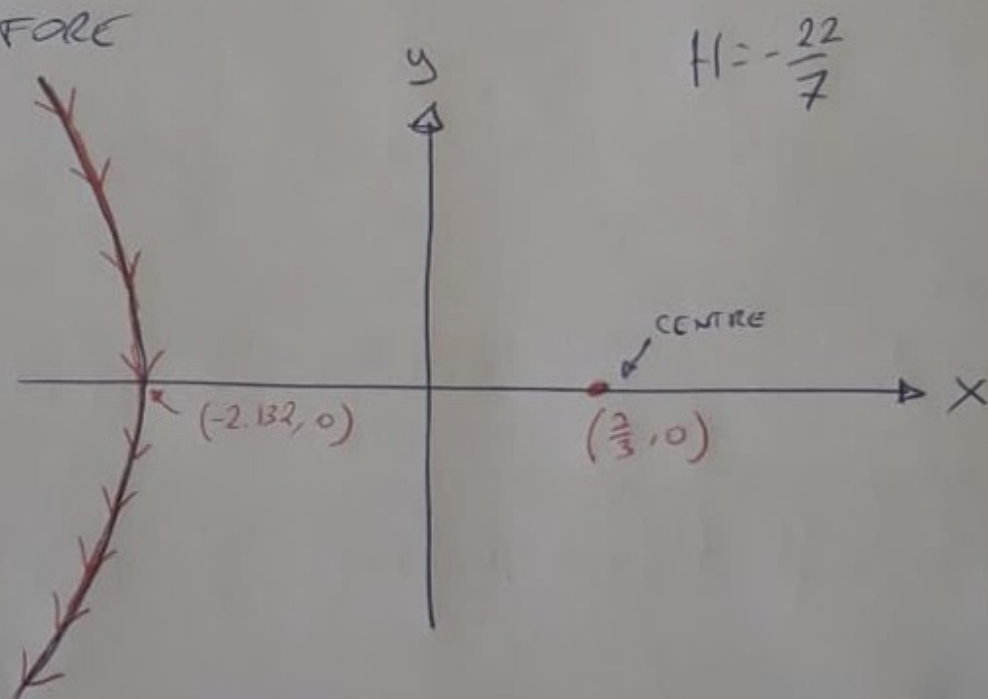
IT TURNS OUT THAT FOR x GREATER THAN THE ROOT $x \approx -2.132$ THERE IS NO VALUE. HENCE THE LEVEL CURVE $H = -\frac{22}{7}$ LIES ALL LEFT THAN THE ROOT

• $x = -2.5 \Rightarrow y \approx \pm \sqrt{8.714} \approx \pm 2.952$

• $x = -3 \Rightarrow y \approx \pm 5.169$

• $x = -4 \Rightarrow y \approx \pm 9.472$

THEREFORE



• ARROW DIRECTION

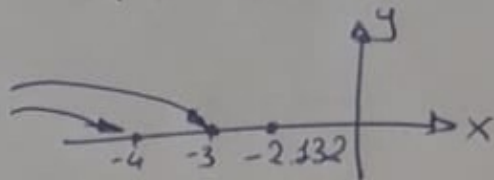
$x = -2.132 \Rightarrow \dot{x} = 0$
 $y = 0 \Rightarrow \dot{y} = -9.504 < 0 \Rightarrow$ ARROW POINTS DOWN

$$I \quad C < -\frac{22}{7} \Rightarrow x^3 + \frac{1}{2}(x^2 + y^2) - 2x < -\frac{22}{7}$$

$$y = \pm \sqrt{-2x^3 - x^2 + 4x + C} \quad \text{with } C < -\frac{22}{7}$$

From the asymptotic analysis we have that $\lim_{x \rightarrow -\infty} H = -\infty$ for fixed y .
Also from II we know that if $y=0$ and $H = -\frac{22}{7}$ there are no real solutions for $x > -2.132$.

SO LET'S FIND
A COUPLE OF
LEVEL CURVES
THAT PASS THROUGH
 $(-3,0)$ AND $(-4,0)$



$$\bullet (x,y) = (-3,0) \Rightarrow H(-3,0) = -\frac{51}{2} < -\frac{22}{7} \quad \checkmark$$

$$H = -\frac{51}{2}$$

PICK SOME VALUES FOR x

$$\bullet x = -4 \Rightarrow y = \pm \sqrt{\frac{141}{2}} \approx \pm 8.336$$

$$\bullet x = -5 \Rightarrow y = \pm \sqrt{\frac{359}{2}} \approx \pm 13.338$$

$$\bullet (x,y) = (-4,0) \Rightarrow H(-4,0) = -48 < -\frac{22}{7} \quad \checkmark$$

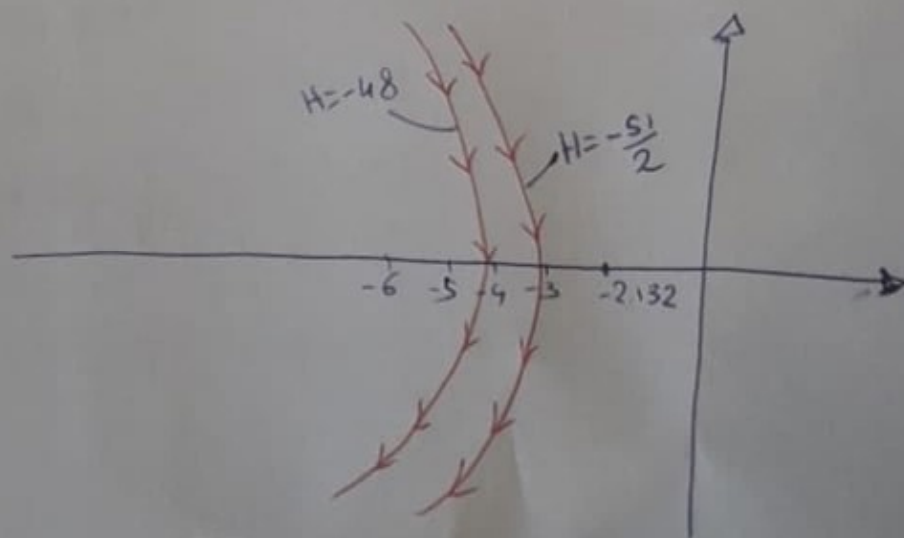
$$H = -48$$

PICK SOME VALUES FOR x

$$\bullet x = -5 \Rightarrow y = \pm \sqrt{109} \approx \pm 10.44$$

$$\bullet x = -6 \Rightarrow y = \pm \sqrt{276} \approx \pm 16.613$$

THEREFORE



IV $C = \frac{3}{2}$

$$x^3 + \frac{1}{2}(x^2 + y^2) - 2x = \frac{3}{2} \Rightarrow y = \pm \sqrt{-2x^3 - x^2 + 4x + 3}$$

• INTERSECTION WITH THE VERTICAL AXIS

$$x=0 \Rightarrow y = \pm \sqrt{3}$$

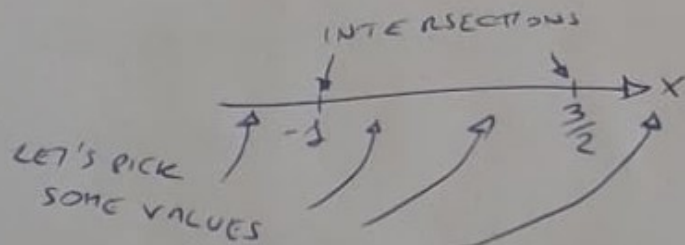
• INTERSECTION WITH THE HORIZONTAL AXIS $y=0$

$$x^3 + \frac{1}{2}x^2 - 2x - \frac{3}{2} = 0$$

One intersection is known: $(-1, 0)$ at the saddle point, hence we can rewrite the equation as (factorizing)

$$(x+1)^2 \left(x - \frac{3}{2}\right) = 0$$

There is another intersection at $(\frac{3}{2}, 0)$



$$x = -2 \Rightarrow y = \pm \sqrt{7} \approx \pm 2.646$$

$$x = -3 \Rightarrow y = \pm \sqrt{36} \approx \pm 6$$

$$x = -4 \Rightarrow y = \pm \sqrt{53} \approx \pm 9.95$$

$$x = 0 \Rightarrow y = \pm \sqrt{3} \approx \pm 1.732$$

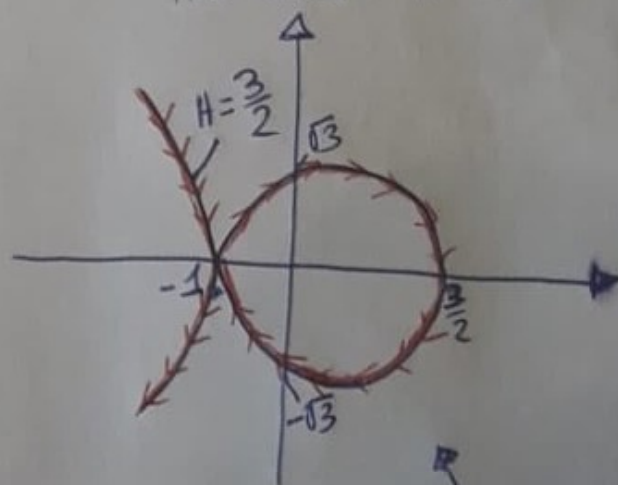
$$x = \frac{1}{2} \Rightarrow y = \pm \sqrt{\frac{9}{2}} \approx \pm 2.121$$

$$x = \frac{2}{3} \Rightarrow y = \pm \sqrt{\frac{125}{27}} \approx \pm 2.151$$

$$x = 1 \Rightarrow y = \pm \sqrt{4} \approx \pm 2$$

$$x = 2 \Rightarrow y = \pm \sqrt{-9} \text{ NO VALUES FOR } x > \frac{3}{2}$$

THEREFORE PLOTTING ALL THE VALUES WE GET



REMEMBER: TO FIND THE ARROW DIRECTION SUB SOME VALUE AND LOOK AT $\dot{x}, \dot{y}$

III $-\frac{22}{7} < C < \frac{3}{2}$

LET'S CHOOSE ~~THE~~ $C=0$ WHICH IS THE SIMPLEST LEVEL CURVE

$$x^3 + \frac{1}{2}(x^2 + y^2) - 2x = 0 \Rightarrow y = \pm \sqrt{-2x^3 - x^2 + 4x}$$

• INTERSECTIONS WITH HORIZONTAL AXIS

$$y=0 \Rightarrow -2x^3 - x^2 + 4x = 0$$

$$-x(2x^2 + x - 4) = 0$$

$$x=0 \vee 2x^2 + x - 4 = 0$$

$$x=0 \vee x = \frac{-1 \pm \sqrt{1+32}}{4} = \frac{-1}{4} \pm \frac{\sqrt{33}}{4} = \begin{cases} 1.186 \\ -1.686 \end{cases}$$

• DOMAIN

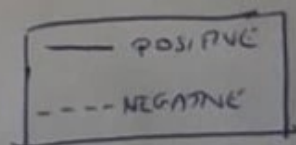
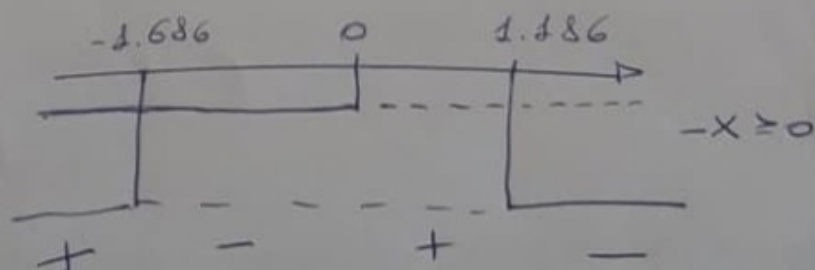
$$y \text{ real} \Leftrightarrow -2x^3 - x^2 + 4x \geq 0$$

$$-x(2x^2 + x - 4) \geq 0$$

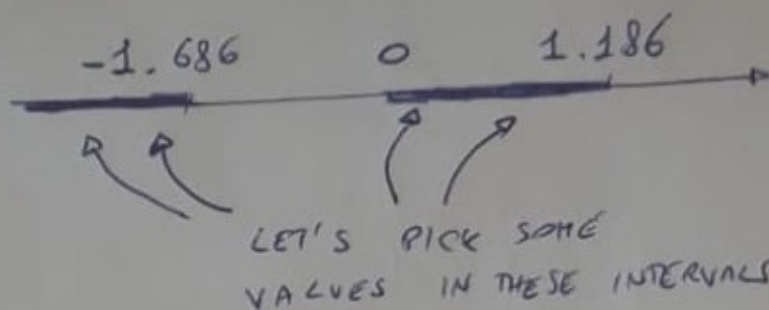
$$-x \geq 0 \text{ if } x \leq 0$$

$$2x^2 + x - 4 \geq 0 \text{ if } x \leq -1.686 \vee x > 1.186$$

THEREFORE



$y = \pm \sqrt{-2x^3 - x^2 + 4x}$ exists only for $x \leq -1.686$ or for $0 \leq x \leq 1.186$



$$x = 0.5 \Rightarrow y \approx \pm 1.225$$

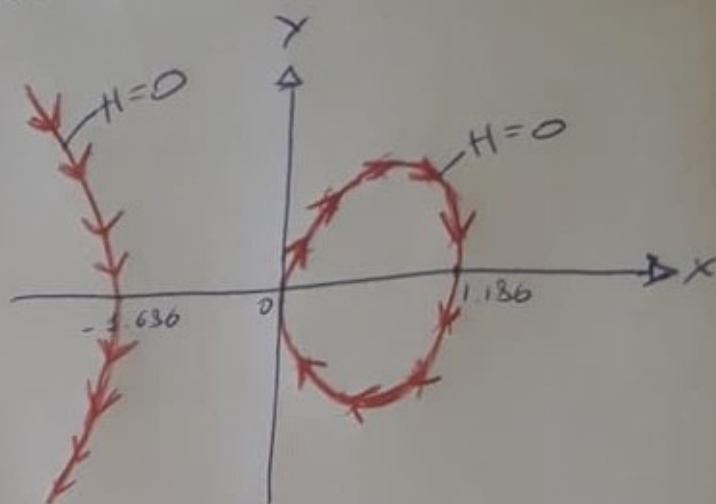
$$x = 1 \Rightarrow y = \pm 1$$

$$x = 0.25 \Rightarrow y \approx \pm 0.352$$

$$x = 0.75 \Rightarrow y \approx \pm 1.594$$

$$x = -2 \Rightarrow y = \pm 2$$

$$x = -2.5 \Rightarrow y \approx \pm 3.873$$



$$\nabla C > \frac{3}{2}$$

Knowing that $(x, y) = (\frac{3}{2}, 0)$ has $H = \frac{3}{2}$ and from the asymptotic analysis $\lim_{x \rightarrow \infty} H = \infty$ for fixed y , let's choose $x = 2, y = 0$

$$H(2, 0) = 2^3 + \frac{2^2}{2} - 2 \times 2 = 8 + 2 - 4 = 6$$

SO LET'S STUDY THE LEVEL CURVE $H = 6$

$$H(x, y) = x^3 + \frac{1}{2}(x^2 + y^2) - 2x = 6$$

$$y = \pm \sqrt{-2x^3 - x^2 + 4x + 12}$$

• INTERSECTIONS WITH HORIZONTAL AXIS, APART FROM $(2, 0)$, IF THEY EXIST.

$$y = 0 \quad -2x^3 - x^2 + 4x + 12 = 0$$

FACTORIZATION (SINCE WE KNOW ONE ROOT ALREADY)

$$-(x-2)(2x^2 + 5x + 6) = 0$$

$$2x^2 + 5x + 6 = 0$$

$$x = \frac{-5 \pm \sqrt{25 - 48}}{4}$$

COMPLEX.
HENCE
 $x = 2$ IS THE
ONLY INTERSECTION
WITH $y = 0$

• INTERSECTION WITH THE VERTICAL AXIS

$$x=0 \Rightarrow y = \pm \sqrt{12}$$

• DOMAIN

$$y \text{ real} \Rightarrow -2x^3 - x^2 + 4x + 12 \geq 0$$

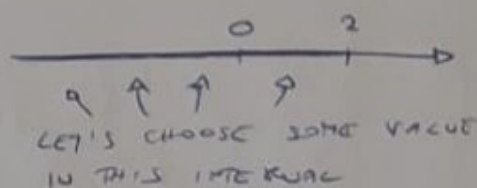
$$-(x-2)(2x^2 + 5x + 6) \geq 0$$

$$-(x-2) \geq 0 \text{ if } x \leq 2$$

$$2x^2 + 5x + 6 \geq 0 \quad \forall x \in \mathbb{R}$$

Hence

$$-2x^3 - x^2 + 4x + 12 \geq 0 \text{ for } x \leq 2$$



$$x = 1.5 \Rightarrow y = \pm 3$$

$$x = 1 \Rightarrow y = \pm \sqrt{13}$$

$$x = 0.5 \Rightarrow y \approx \pm 3.674$$

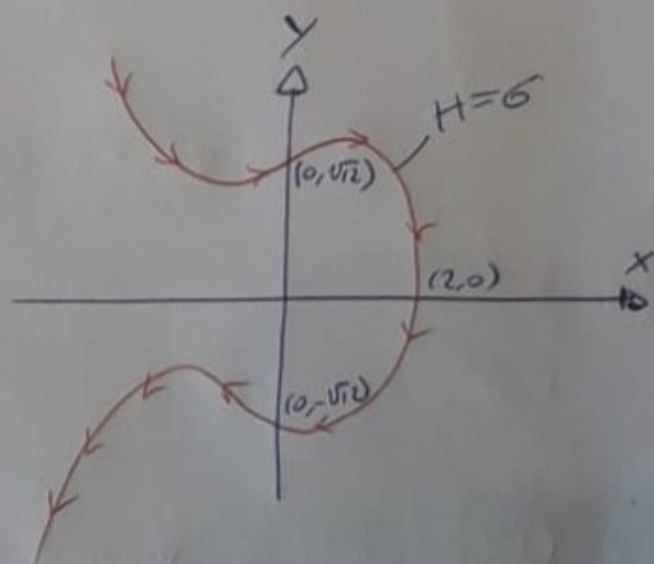
$$x = -0.5 \Rightarrow y \approx \pm 3.162$$

$$x = -1 \Rightarrow y = \pm 3$$

$$x = -2 \Rightarrow y = \pm 4$$

$$x = -4 \Rightarrow y = \pm 6\sqrt{3}$$

• LET'S PLOT THESE



FINALLY WE CAN PUT IT ALL TOGETHER

